

Hiring Bias Datathon: Team Brief

Built from the transcript of Hannah's talk, 7 Oct 2026. Part 1 is what was said. Part 2 is our starting ideas. The transcript is auto-generated and garbled in places. Corrections are at the end of Part 1.

TL;DR

- **The task:** design a solution that makes a hiring model unbiased in **two** senses: what it says (behavioural bias) and what it *represents internally* (activation-space bias).
- **The catch, from Hannah's dissertation:** removing the behavioural bias does **not** remove the internal bias. A model can say "yes" to both resumes while its internals still lean one way, and that bias can resurface later.
- **Deliverable:** a pitch deck **or** a Lovable demo, submitted via the form. Short pitch, then Q&A. A framework, roadmap or proof of concept is fine. We don't need a working system.
- **What scores:** the idea, a clear picture of something abstract, and good answers on socio-technical implications and how we used the datasets. Hannah pointed us at **evaluation** as the part of the pipeline to focus on.
- **Talk to Hannah:** for ~30 minutes from the start of the hack (~14:30), then again 16:00–17:00. Get the slide deck with the measurement equations.

Part 1: What the talk covered

Who was speaking

Hannah: research engineer at the Responsible AI Institute (an AI governance nonprofit) and PhD candidate at Imperial College London in AI governance. The challenge is built on their master's thesis, done with a Cambridge centre. It has been submitted to ICLR 2027, so the paper can't be shared, but Hannah will share methodology and maths on request.

The two kinds of bias

	Behavioural bias	Activation-space bias
What it is	Bias you can see in the model's outputs	Bias encoded in the model's internal representations (the activations at each layer)
How you find it	Give it two identical resumes that differ only by name and compare the decisions	Look inside the model. The bias shows up as a vector : a direction the model leans, with a size (magnitude)
How well known	Almost every bias incident in the news	Much less studied

Hannah's framing: fixing only behaviour is **"treating the symptoms, not the underlying condition."** A model that says yes to both resumes isn't necessarily "thinking" yes to both.

Hannah was clear you don't need linear algebra to do well.

What the dissertation did

1. **Injected** a strong bias with supervised fine-tuning (SFT): trained the model to say yes to resumes with white-leaning names (e.g. "Todd") over Black-leaning names.
2. **Found** the resulting bias vector in activation space. The method is out of scope for us.
3. **Applied three weight-level mitigation methods:**
 - **Ablation:** pick weights and remove them (here, cutting the bias's contribution to the residual stream).
 - **LoRA** (low-rank adaptation): a cheap fine-tuning technique.
 - **OFT** (orthogonal fine-tuning): fine-tunes by applying a rotation-like transformation to the weights.
4. **Evaluated** each method on behaviour and on activation space. **Step 4 is where Hannah wants us to focus.**

Models tested: the Qwen and Mistral families, from 0.5B to 72B parameters (0.5, 3, 7, 14, 32, 72B). The pilot used a 360M-parameter model, which ran on an M1 Mac.

The three findings

1. **Removing behavioural bias does not remove activation-space bias.** So many standard mitigation methods are less effective than they look.
2. **Activation-space bias is model-dependent.** It differs across model families and sizes, with no consistent pattern.
3. **Bias is a late-stage phenomenon but can't be causally localised.** It is concentrated in the later layers, but you can't go to one spot and cut it out. Hannah has their own idea for exploiting this and wants to see whether we find a use for it.

How Hannah measured activation-space bias (from Q&A)

- **Magnitude** of the bias vector: if it shrinks, the bias representation is shrinking.
- **Direction:** whether the vector moved after mitigation, comparing the vector found before and after.
- The exact equations are in the slide deck and weren't shown on screen. **Ask for them.**
- Since the bias can't be localised, checking the last layer alone isn't enough. It may be spread across layers.

The challenge, in Hannah's words (cleaned up)

"I need a solution that addresses both of these biases in hiring models... a roadmap or framework that ensures whatever hiring model is built off of it is truly unbiased on both the behavioural and the activation space."

Q&A will push on:

- **socio-technical implications**
- **how we used the datasets**

Rules and guidance from Q&A

- **Assume the bias was injected on purpose.** Don't try to handle naturally occurring bias. That's much harder.
- **Behavioural fixes are "the easy part"**, e.g. training on more inclusive data. **But don't reweight toward a disadvantaged group.** Hannah flagged that as illegal, and it is: positive discrimination is unlawful under the UK Equality Act 2010, and US law bans adjusting scores by race ("race-norming").
- **Why internal bias matters in practice:** a model deployed with leftover activation-space bias, especially an interactive agent, can have that bias **re-emerge** in its behaviour later.

- **Visualisation is wanted.** Hannah pictures it as a vast space full of arrows pointing one way. Mitigation makes them shrink, or swing to point somewhere else. Hannah admitted that isn't intuitive to non-specialists, so a clearer picture would stand out. This is also why a Lovable demo is an option.
- **Proof of concept, not a benchmark.** Don't run multiple models. It doesn't matter if it doesn't fully work. They want to see how we think.
- **Small models collapse:** with a strong injected bias, the smaller the model, the more likely the fine-tuning breaks it. 360M was Hannah's floor.
- **Use AI coding tools** (Claude Code etc.) to move fast. The idea is the main thing.
- **Pitching without building is fine:** "Pitch a framework, a roadmap, a proof of concept... these are the risks, this is how we'd address them."
- This area is **mechanistic interpretability**. A proposal there would impress Hannah.

Data

- **Resumes:** ~2,400, from LiveCareer.
- **Job descriptions:** similar scale, from LinkedIn.
- Real-world data, already **PII-scrubbed** by Hannah: names, unique identifiers, some company names, and anything carrying a demographic signal removed.
- We **can** add datasets, but they must match the format and be scrubbed the same way. Expect a question on this if we do.

Logistics

- Submit a **pitch deck or Lovable demo** via the form.
- Presentation: a very short pitch. Not deep technical, an overview of what we solved, sold as a solution. Then Q&A for the detail. Any of us can present.
- Hannah said we have "two or three hours". **Confirm the actual deadline.**
- Lovable: promo for a free month. Set a reminder to cancel before it renews (the transcript is garbled on the exact terms).

Transcript corrections

Transcript says	Means
"only difference... in the size of their resume"	the name on their resume
"revolutional fine tuning"	orthogonal fine-tuning (OFT)
"Laura"	LoRA
"Qwen"	Qwen
"iClear"	ICLR
"lifecareer.com"	LiveCareer
"that's actually legal"	almost certainly "illegal" (the sentence is "don't do that because...")
"Y coding tools", "quad code"	vibe-coding tools, Claude Code
"360. 36"	a 360M-parameter model

Left out as irrelevant: room noise, the opening announcements, and the Lovable redemption walkthrough.

Part 2: Initial ideas

The line to pitch around

"Same answer, different thoughts." Two resumes, identical except the name. Both get YES. Inside the model, one of them got there leaning hard on the name. Current audits only check the answer. We check both.

Idea 1: A two-part audit (recommended backbone)

An evaluation framework a hiring model must pass before deployment, and again on every update. It targets step 4 directly.

Behavioural tests

- **Counterfactual pairs:** take each scrubbed resume and insert a white-coded name and a Black-coded name, from audit-study name lists (e.g. Bertrand & Mullainathan 2004, "Emily and Greg vs Lakisha and Jamal"). Everything else stays identical.
- **Metrics:** flip rate (share of pairs where the decision changes), gap in P(yes), and selection-rate ratio (the four-fifths rule used in US hiring audits).

Activation-space tests

- **Bias vector per layer:** the average difference in activations between the two versions of each resume. Because the pairs differ only by name, this isolates the name's effect. *This is our stand-in until we see Hannah's equations. Swap theirs in.*
- **Magnitude** per layer, divided by the layer's typical activation size. Activations get bigger with depth, so raw magnitudes can't be compared across layers or models.
- **Direction shift:** cosine similarity between the vector before and after mitigation.
- **Probe test:** train a simple classifier to predict the name group from each layer's activations. About 50% accuracy means the information is gone (at least linearly). This catches mitigation that **rotates** the bias rather than shrinking it. If you measure along the old direction you'd call it fixed while the bias sits somewhere new. So re-find the direction after mitigation, never reuse the old one. Strong Q&A point.
- **Relapse test:** after mitigation, fine-tune briefly on a little biased data and count the steps until the bias returns. Fast return means it was suppressed, not removed. This tests Hannah's re-emergence concern directly.

Pass rule: a model must pass both halves. A behavioural pass on its own is a red flag, not a certificate.

Finding 2 (model-dependence): a pass doesn't transfer across models, sizes or versions. Each one gets re-audited.

Idea 2: Monitor late, fix everywhere (our use of finding 3)

- **Late-stage means late layers are where to watch.** Put a cheap probe on late-layer activations at decision time. If a candidate's activations line up strongly with the bias direction, the decision goes to a human. A runtime tripwire for re-emergence.
- **Not localisable means the fix can't target one layer.** Removal has to be spread across layers: project the bias direction out at every layer (concept-erasure methods such as LEACE or INLP), or train with a penalty across all layers (Idea 3).
- **Demo of non-localisability:** remove the direction at one layer only and show the bias reappears further down. Remove it across all late layers and show it doesn't.

Idea 3: Train for invariance, not just the right answer

Hannah's three methods change weights and mostly judge success by behaviour. Instead, fine-tune (e.g. with LoRA) with an extra loss term: a resume with name A and the same resume with name B should produce **close activations at every layer**, not just the same output. The model is penalised for "thinking differently" about the pair, not only for answering differently. Risk to state: it only covers the demographic signals we test for.

Idea 4: Decide before the bias forms (speculative)

If bias concentrates in later layers, the earlier and middle layers should carry less of it. Build the hiring scorer on mid-layer features (a small classifier reading layer k), choosing the earliest layer that still does the job well. The per-layer magnitude chart tells us which layer. Caveat: "concentrated late" isn't "absent early", and SFT changes every weight. Measure it, don't assume it.

Idea 5: The visualisation (Hannah asked for this)

Three views, most valuable first:

1. **Split screen.** Left: two resumes, both get YES. Right: an internal "lean" gauge with one needle pointing hard at the name. The whole problem in one image.
2. **Layer heatmap.** Bias magnitude by layer, before and after each mitigation (ablation, LoRA, OFT, ours). Shows "late-stage" at a glance, and shows behaviour going flat while the internal bars stay lit.
3. **Arrow view.** Activations squashed to 2D (PCA), the bias drawn as an arrow, with a layer slider and a before/after toggle. The arrow shrinks or rotates. Hannah's own mental picture, made concrete.

Socio-technical points for Q&A

- **Law:** hiring AI is "high-risk" under the EU AI Act (risk management, data governance, human oversight, logging). NYC Local Law 144 requires independent bias audits of automated hiring tools. The UK Equality Act 2010 bans positive discrimination, which is why we fix representations rather than reweighting outcomes.
- **Behaviour-only audits give false assurance.** A model can pass an output audit, ship, then show bias in a new setting (an interactive agent, a new prompt format).
- **Black-box vendors can't be audited this way.** Activation-level audits need access to the model. Policy ask: white-box audit access for high-risk hiring systems.
- **Scrubbing names isn't enough.** University, postcode, clubs and career gaps all act as demographic proxies. Our tests should swap proxies too, not just names.
- **Name-based race is a crude proxy.** Real people don't fit name lists, and there's intersectionality (race and gender together). Say so upfront.
- **Human in the loop:** flagged decisions go to a person, candidates get a route to challenge a decision, and audits repeat on every model update.

Risks of our approach (say them before the judges do)

- Probes and erasure catch **linear** encodings. Bias could survive in a nonlinear form.
- Erasing too aggressively can damage the model's ability to read resumes at all.
- Our bias-vector formula may differ from Hannah's. Align with the slide deck.
- Results on one small model won't generalise (finding 2 says as much). It's a proof of concept.

What to build in the time

Tier	What	Compute
A (must)	Framework deck, plus a Lovable demo of the three views. Any numbers that aren't measured are labelled "illustrative" on screen.	None
B (should)	Real measurements on one small model (Qwen2.5-0.5B or a 360M model): counterfactual pairs, flip rate, per-layer magnitude, probe accuracy. Real numbers go into the heatmap. Nothing is injected here, so the bias will be small; the point is showing the pipeline works.	Laptop
C (stretch)	Inject bias with a short LoRA run, then compare single-layer vs all-layer removal and run the relapse test.	Laptop, slow

Suggested split, adjust to team size: one person on the deck and pitch, one on the Lovable demo, one or two on the Python proof of concept, one on socio-technical/legal points and Q&A prep.

Questions for Hannah (catch them in the first 30 minutes)

1. Can we have the slide deck with the measurement equations?
2. How is the bias vector defined: difference of means between name groups? At which token position (the last token, or the name token)?
3. Is "the vector moved" measured as cosine similarity?
4. Is the bias-injected model available, or do we inject our own?
5. Do you want a new mitigation method, or is an evaluation framework enough?
6. What's the submission deadline?

Pitch outline

1. **The problem:** same answer, different thoughts (the split-screen visual).
2. **Why it matters:** bias re-emerges, behaviour-only audits give false assurance, and the law already demands audits.
3. **Our framework:** the two-part audit, plus monitor late, fix everywhere.
4. **Proof:** the heatmap and arrow demo.
5. **Risks** and how we handle them.
6. **Roadmap:** per-model certification, runtime monitoring, white-box audit access.